A COVID-19 Case Study on Model Risk and Diversification Limits in a Fifteen-Stock Portfolio

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Introduction

The COVID-19 pandemic produced one of the sharpest, most synchronized shocks in recent financial history. This report uses standard risk and dependence measures employed in heavy-tailed multivariate modeling on real daily price data for fifteen US stocks, JAN 2019 to DEC 2022.

The fifteen names were originally going to be chosen to mirror US GDP-weighted sector shares, but such a basket could mostly show every stock behaving alike, which defeats the purpose of contrast. The cohort ended up being chosen to span sectors that lived through the pandemic differently, be that sharp crashes (airlines, aerospace, energy, real estate), stay-at-home surges (streaming, e-commerce), or defensives (utilities, staples, healthcare); within two sectors, a stable name is paired with a high-beta one (Microsoft/Nvidia; Pfizer/Regeneron) so that the later dependence analysis can differentiate shared-sector effects from those of shared-volatility. Every name was already listed publicly well before JAN 2019, giving each a period of ordinary trading behavior against which to measure market crash.

The report follows a line of inquiry from data to risk. The following section covers exploratory data analysis and cleaning procedures. Next, the report explores whether daily returns behave as the normal model predicts, alone and jointly. The report then estimates the hypothetical portfolio's risk ($1,000 per stock, held fixed) in three different ways and backtests each against a subsequent, more volatile, out-of-sample year. Lastly, the report assesses whether the dependence *between* stocks is Gaussian and what that implies for a simultaneous crash. Methodological choices are justified where they arise rather than collected into a separate section as the reasoning differs by task.

Data

Daily prices for the fifteen stocks were obtained from Yahoo Finance via R's quantmod package, spanning 02 JAN 2019 to 30 DEC 2022 (1008 trading days)---Yahoo was chosen for its free, daily history and, particularly, its adjusted-close series that incorporates the effect of stock splits and dividend payments, relevant given that three of the fifteen names underwent such a corporate action within the sample window. As it turns out, Yahoo periodically revises past adjusted-close figures as new dividends are declared, so a raw snapshot of the pulled data was saved to disk immediately after download, every subsequent step in the analysis loading from that copy; this ensures that the results reported here won't change if the price series is later revised.

Table 1 lists the fifteen tickers. Sectors that behaved differently through the pandemic are represented, and two sectors (technology, healthcare) each pair a low-volatility stock against a high-beta one so that within-sector dependence isn't confounded with within-sector homogeneity; that is to say, two similarly volatile stocks moving together would only really indicate that the sector moves together whereas a pair of stable and high-beta stocks separates shared sector exposure from each stock's own risk behavior for the dependence analysis in the end of the report. Each stock has a ~14-month pre-pandemic baseline, a constraint that itself ruled out several candidate stocks that had listed too recently to supply in-sample returns.

Ticker	**Sector**	**Pandemic role**
WMT	Consumer staples	defensive
UAL	Airlines	deep crash, slow recovery
NFLX	Media	surge
ETSY	Consumer discretionary	surge
XOM	Energy	crash
PFE	Healthcare (pharma)	defensive
REGN	Healthcare (biotech)	surge
JPM	Banking	crash, then recovery
MSFT	Tech	stable
BA	Aerospace/defense	deep crash
DUK	Utilities	defensive
DIS	Media/discretionary entertainment	divergent
TSLA	Autos	surge
NVDA	Semiconductors	crash, then boom
SPG	Real estate	crash

Table 1*: fifteen-stock portfolio.*

Each stock's series carries daily open, high, low, close, adjusted close, and volume, with only adjusted close used in the report. The sample is split into an in-sample estimation window (02 JAN 2019 to 31 DEC 2021, 756 trading days) and a held-out backtest year (2022, 251 trading days). The in-sample window covers the pre-pandemic baseline, the crash, and the initial recovery, the backtest window a new stress period driven by interest-rate rises coupled with a resurgence in COVID-19 cases, against which the risk models below are tested. No stock had a missing observation anywhere in the sample, though three corporate actions required cleaning. TSLA split 5-for-1 on 31 AUG 2020 and 3-for-1 on 25 AUG 2022; the second split, confirmed via Yahoo's corporate actions record, occurred during the backtest year. NVDA split 4-for-1 on 20 JUL 2021. SPG cut its quarterly dividend from $2.10 to $1.30 (a ~38.1% reduction) between its FEB and JUL 2020 payments, preceding a gradual recovery to $1.80 by the end of 2022. Adjusted close already accounts for split and dividend adjustments, so none of the above required manual cleaning; querying Yahoo's corporate actions record for each stock, given the price series alone removes the discontinuity and thus wouldn't have revealed a split, is what caught the second TSLA split to begin with. A small number of observations showed a daily log return of zero, meaning the price was identical to that of the previous day—naturally, this typically stems from a stale (low trading) day rather than zero volatility; left uncorrected, this would understate the kurtosis estimate and misrepresent the tail behavior examined below. These cells (36 in total, ~0.238% of the panel, spread across 35 distinct trading days) were marked as missing as opposed to dropping the whole day; indeed, a stale print for one stock says nothing about whether the other fourteen traded as per usual on that given day. Affected dates were also checked using the extreme-move list below, and none coincide.

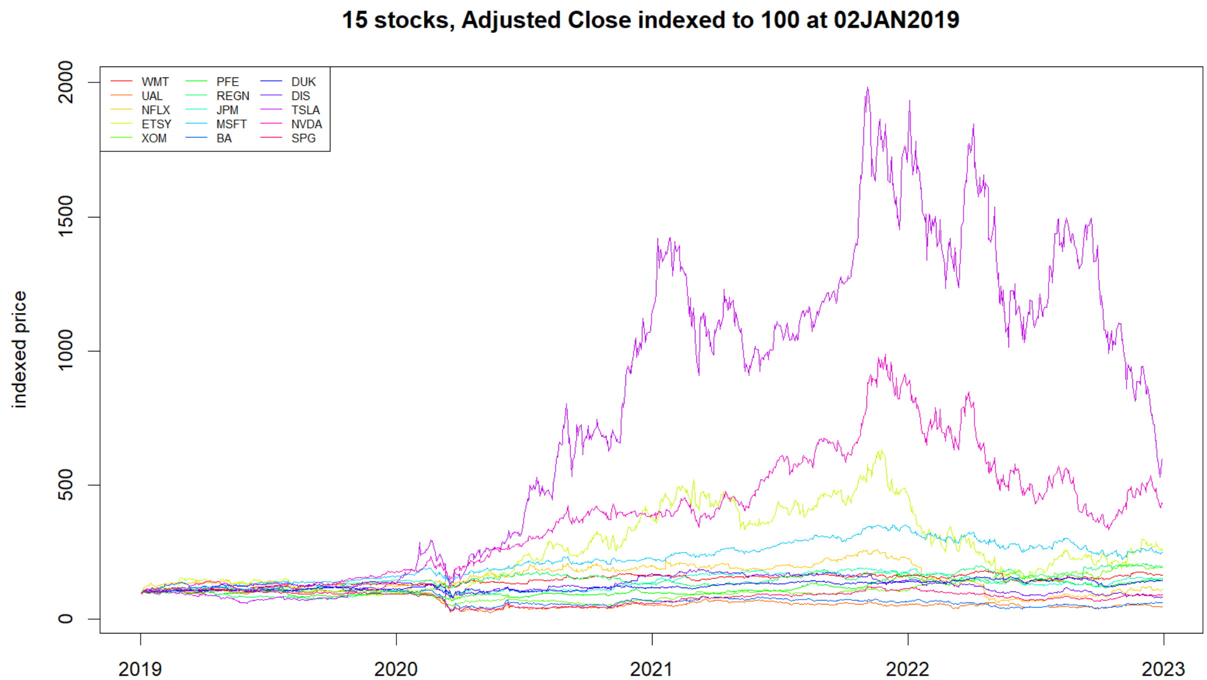


Figure 1

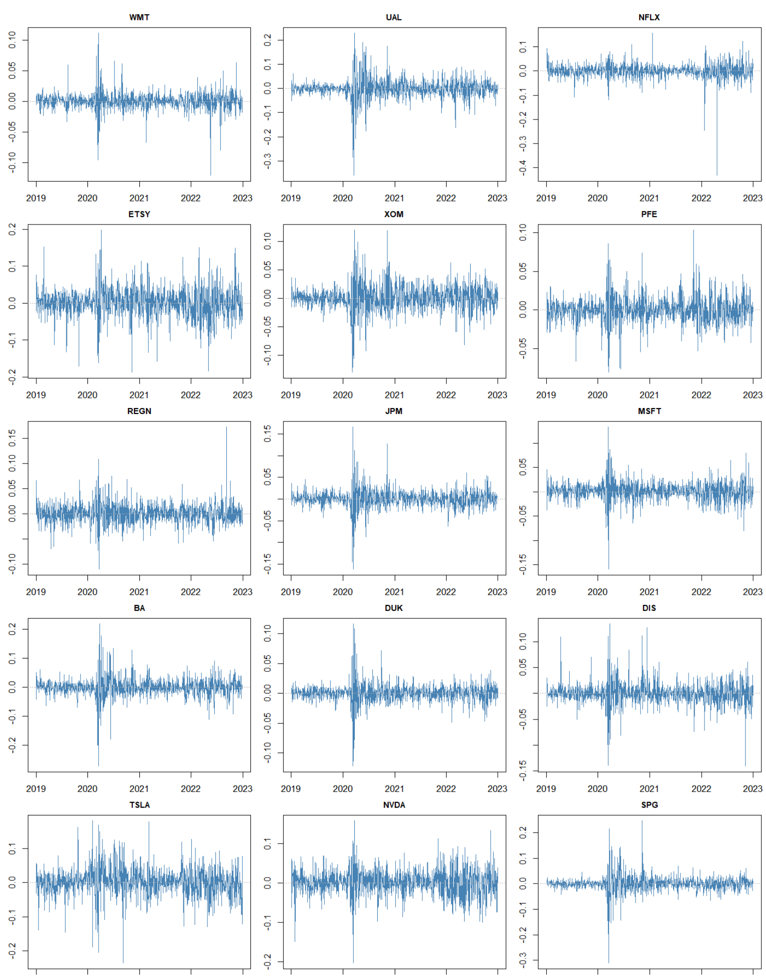


Figure 2

Before running any tests, the cleaned panel was inspected visually. **Figure 1** shows an indexed price chart, **Figure 2** a small multiples plot of daily log returns for all of the stocks, and these were used to check for anomalies that summary statistics could've missed such as a flat stretch, a spike seemingly out of proportion, etc. Both confirm that the portfolio behaved within reason and, moreover, provide immediate visual evidence of the pandemic crash as a shared, near-simultaneous event in FEB/MAR 2020. This is formally addressed in the following section.

As a precaution, every daily log return exceeding 0.15 in absolute magnitude was flagged; this threshold was chosen to sit several standard deviations above typical daily volatility even for the portfolio's most volatile stock (TSLA, ~4.23% daily) so that only such exceptional days are caught. Fifty-two of these days were identified across the stocks and form the starting point for the discussion below.

Returns and Normality

Date	Ticker	Log return	Event
16 MAR 2020	SPG	-31.08%	FEB/MAR 2020 sell-off amid global COVID-19 disruption
18 MAR 2020	UAL	-36.08%	Same episode
09 NOV 2020	SPG	24.58%	Vaccine efficacy announcement induced market-wide rally
21 JAN 2022	NFLX	-24.58%	Weak subscriber-growth guidance
20 APR 2022	NFLX	-43.26%	First subscriber loss in ten years (largest single move)

***Table 2** selects several of the most illustrative of the fifty-two extreme days identified in the Data section, tied to the events that offer plausible explanations.*

Most of the fifty-two days cluster in the weeks surrounding the FEB/MAR 2020 crash, UAL alone accounting for twelve of those days. That it stretches as late as NOV 2020 is evidence that uncertainty about the shape and timing of a travel recovery persisted well beyond the market's most acute stress, an expected outcome during an international viral pandemic. XOM is not featured because its widely-reported APR 2020 dip, when crude futures traded negative, never produced a *single-day* equity move beyond 0.15—in essence, a futures contract's price is different from the underlying company's equity value. Furthermore, not every extreme move necessarily traces to the pandemic: REGN's own largest move (17.27%, 08 SEP 2022) lies beyond the pandemic's acute period and thus has no pandemic-related catalyst. NFLX's 2022 drop, on the other hand, ties back to the pandemic, even though this happened well after the initial crash—subscriber growth that had been supercharged during quarantines reversed sharply as the world reopened.

Table 2 shows large swings, though assessing normality formally requires skewness (asymmetry, equal to 0 for a normal distribution) and kurtosis (heaviness of a distribution's tails and how sharply it peaks relative to a normal distribution), which is reported here as excess kurtosis so that a normal distribution again corresponds to a measure of 0. The Jarque-Bera test combines both into a univariate hypothesis test of normality.

Ticker	Skewness	Excess kurtosis	JB Stat
NFLX	-2.86	38.86	68,321
SPG	-0.91	20.31	17,456
WMT	-0.20	13.90	8,157
DUK	-0.20	13.69	7,878
UAL	-0.72	12.54	6,695
JPM	-0.04	11.84	5,897
BA	-0.40	11.47	5,559
DIS	0.12	8.37	2,954
MSFT	-0.32	7.71	2,513
REGN	0.61	6.32	1,748
XOM	-0.19	4.52	870
PFE	0.16	4.16	727
TSLA	-0.26	3.65	574
NVDA	-0.25	2.77	335
ETSY	-0.22	2.77	331

***Table 3**: Skewness, excess kurtosis, and the Jarque-Bera test, all stocks (sorted by excess kurtosis; every test rejects normality at p < 0.001).*

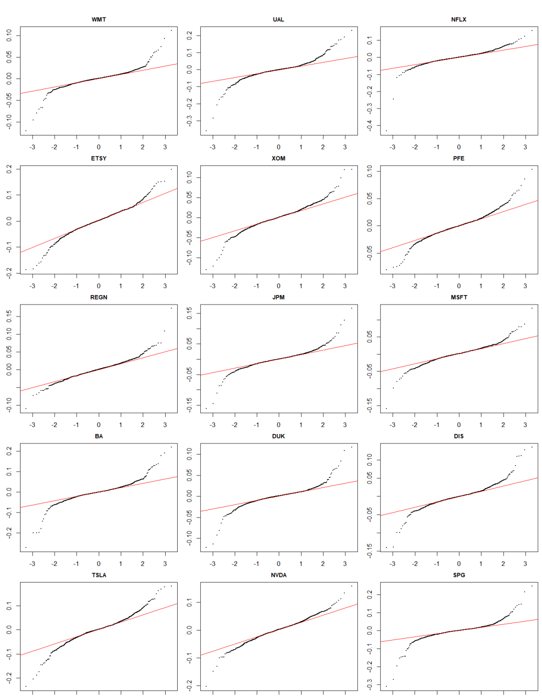
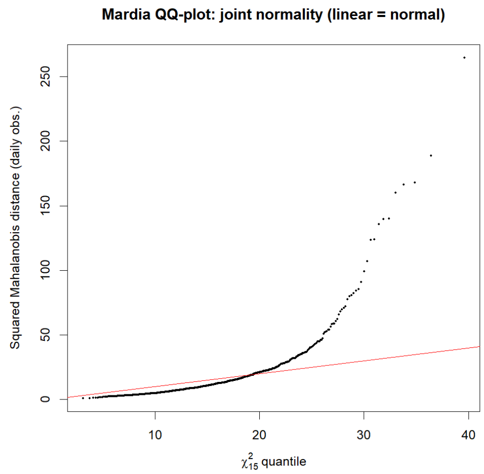


***Figure 3** shows quantile-quantile plots against a normal reference for the fifteen stocks, nearly every panel exhibiting the inverted-s curve characteristic of heavier-than-normal tails, corroborating **Table 3**.*

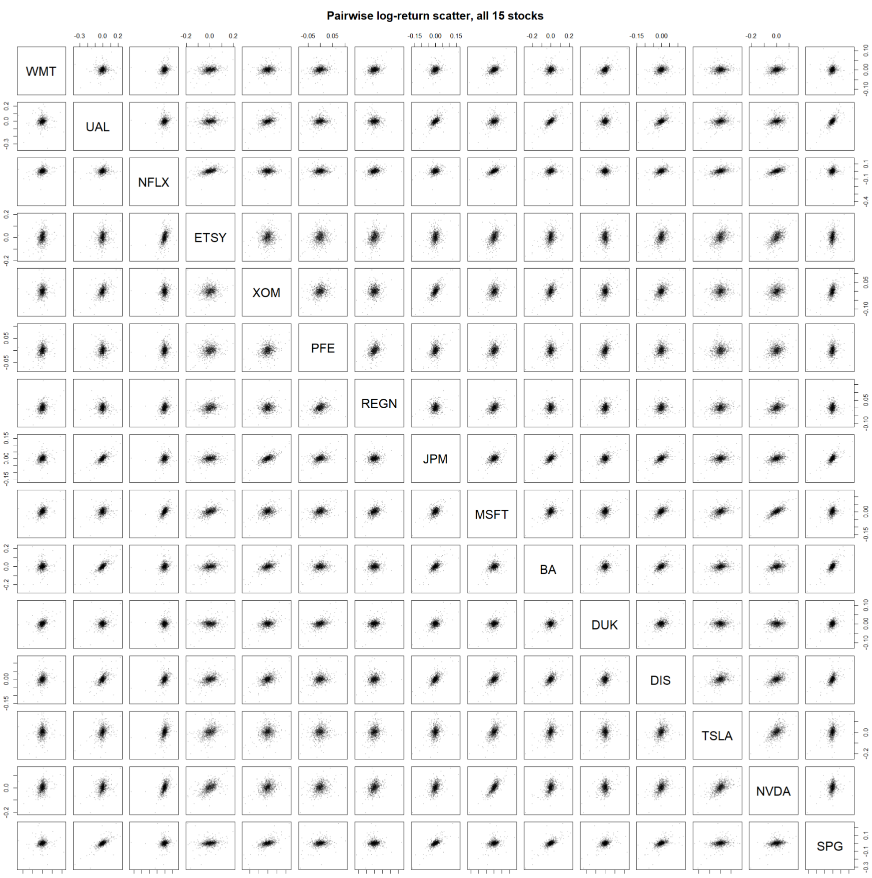
All fifteen tests reject normality decisively, so, considered *individually*, the normal model does not hold for the stocks' returns. NFLX's excess kurtosis is nearly twice that of the next-highest-kurtosis stock, yet this can be explained by the two days already identified in **Table 2**, demonstrating that a small number of extreme observations can strongly drive a fourth moment statistic. NVDA, by contrast, is the closest to matching actual Gaussian quantiles (ratios span 0.97–1.07). It is worth note that ETSY's excess kurtosis is marginally lower, yet it ties SPG for the largest deviation at 99% (1.29).

Comparing each stock's actual quantiles against those implied by a normal distribution with the same variance shows that the moderate 5th and 95th percentiles are typically thinner than the normal model predicts whereas positive excess kurtosis reveals that the extreme 1st and 99th percentiles are fatter. SPG, for example, has 5th and 95th percentiles 0.76 and 0.75 times, respectively, as extreme as a variance-matched Gaussian would imply, its 1st and 99th 1.29 and 1.25 times as extreme. Meanwhile, NVDA's own four ratios being close to 1 is consistent with its low kurtosis in **Table 3**. The univariate normal model is rejected for every stock in the sample; a natural extension checks whether joint, multivariate normality holds across the sample. A defining property of the multivariate normal distribution is that every one of its marginal distributions must itself be normal, so the rejections in **Table 3** conveniently rule this out on their own. Nonetheless, Mardia's test confirms this formally and, with more insight, quantifies it in extending skewness and kurtosis to multiple dimensions.



Here, $n = 972$ reflecting the small number of days removed by the cleaning described earlier, and both the multivariate skewness statistic (11,054.92 on 680 degrees of freedom) and kurtosis statistic (a standardized score of 256.21) reject joint normality ($p < 0.001$ for both). Under multivariate normality, the kurtosis should average $d(d + 2) = 15*17 = 255$, but this is well below the observed value of 626.17. **Figure 4** shows this graphically, plotting each day's squared Mahalanobis distance (a measure of how far that day's fifteen-stock return vector sits from the sample average) in units that account for how the stocks move together against the distribution this quantity should follow under multivariate normality. Put rather simply, the multivariate QQ-plot shows that departure from normality is most visible in the upper tail. Hence, the most extreme days behave more extremely relative to a typical day than a joint Gaussian would predict.

Figure 4



Shifting focus to how the stocks in the sample relate to one another calls for the use of pairwise scatter plots of daily log returns (**Figures 5** and **6**) along with the Pearson correlation matrix computed across the series. **Figure 5** plots all (15*14) / 2 = 105 pairs at once; it is important to note that at this density individual points cannot be read off, but the *overall shapes* are informative and themselves warrant further analysis of the interrelations at play—in any case, prepare to zoom in. Consider, several pairs show a visibly elliptical cloud with points falling loosely along a line, consistent with meaningful positive correlation, whereas others show a more diffused scatter with no discernible direction.

Figure 5

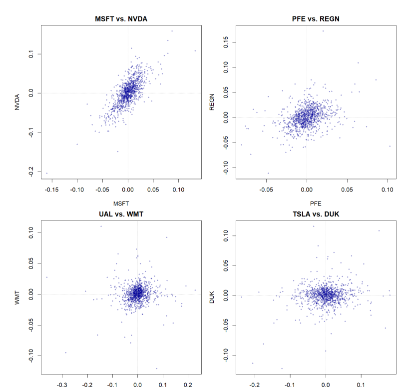


Figure 6

Figure 6 isolates the pairs of same-volatility and stable/high-beta introduced earlier specifically for this contrast—MSFT and NVDA show points falling clearly along a line, PFE and REGN a similar but looser positive pattern, UAL/WMT and TSLA/DUK show a far less pattern-indicative scattering. **Table 4** summarizes the full correlation matrix (reported in full in the accompanying .csv files) by its most- and least-related pairs. The most-correlated list constitutes the stocks that fell hardest *together* in MAR 2020 namely UAL, BA, and SPG. REGN's highest correlation with any other stock in the sample is only 0.43, its correlation with UAL exactly zero, making it the most consistently independent stock on an ordinary trading day; whether this holds during a shared crash requires further analysis using copulas, which is addressed in the end of the report.

Most-related	r	least-related	r
MSFT/NVDA	0.74	UAL/REGN	0.00
UAL/BA	0.72	WMT/UAL	0.07
UAL/SPG	0.69	REGN/SPG	0.08

***Table 4**: Most- and least-correlated pairs (Pearson)*

The four targeted pairs' own correlations (i.e., MSFT/NVDA, PFE/REGN, UAL/WMT, TSLA/DUK) confirm the visual impression from **Figure 6** as well as the intentional pairing described in the exploratory data analysis. Indeed, the within-sector, stable/high-beta pairings move together meaningfully, while the cross-sector stable/crash pairings show weaker average association.

However revealing this might seem, Pearson correlation is a linear measure and can be disproportionately influenced by a small number of extreme-magnitude days which is obviously a risk when working with volatile constituents. **Table 5** recomputes the four targeted pairs' relationships using two rank-based alternatives, Spearman's ρ and Kendall's τ, to mitigate sensitivity to extreme outliers and thus check whether Pearson is in fact being skewed by a relatively small number of large moves.

Pair	Pearson	Spearman	Kendall
MSFT/NVDA	0.74	0.69	0.51
PFE/REGN	0.35	0.38	0.27
UAL/WMT	0.07	0.15	0.10
TSLA/DUK	0.15	0.04	0.03

***Table 5**: Pearson, Spearman, Kendall correlations for four pairs*

TSLA/DUK is an exception across the three measures, instructively so as its Pearson correlation is noticeably higher than either rank-based measure. This can be explained by the multitude of TSLA's own single-day moves being large enough to inflate a linear-correlation measure even though the two stocks' typical daily rankings show little-to-no relationship.

Thus, the logarithmic return series are related to one another albeit unevenly. One consequence is that this motivates testing a heavy-tailed alternative to the normal model when estimating portfolio risk in the next section. Additionally, a correlation figure like Pearson or one of rank-based measures is by construction a summary of typical relatedness that doesn't in fact inform on whether two stocks are more likely to drop together specifically during a crash, motivating the use of copulas in the final stages of the report.

Portfolio Risk and Backtesting

A hypothetical portfolio invests \$1,000 in each of the fifteen stocks at the first price in the sample and holds share counts fixed for all four years; its value on any subsequent day is said fixed holdings valued at that day's price such that day-to-day changes just reflect price movement. Again, dividends are reinvested because adjusted close is used. Daily loss, $L(t)$, is defined as the negative of the daily change in portfolio value. Two standard risk measures are estimated at the confidence levels, 95% and 99%, chosen to capture both a fairly "ordinary" bad day and a genuinely rare, severe one. Value-at-Risk (VaR) at level α is the loss level exceeded with probability $1–\alpha$, Expected Shortfall (ES) the average loss on those days when VaR is exceeded, capturing how bad a bad day that day is expected to be. Both are estimated three-fold from the in-sample loss series (2019–2021). The first estimation is done empirically with the distribution of past losses, no assumption(s) about said distribution. Then, estimation is done using a Gaussian, frankly a naive baseline, given the previous analysis rejects normality for the fifteen stocks individually and jointly; the following checks whether the same conclusion holds once the portfolio is aggregated. Lastly, estimation is done with a Student-*t* distribution fitted directly to the loss series, motivated not least by the heavy tails documented in the previous section.

model	α	VaR	ES
Empricial	95%	\$764	\$1,329
Empirical	99%	\$1,763	\$1,998
Normal	95%	\$798	\$1,013
Normal	99%	\$1,148	\$1,322
Student-*t*	95%	\$805	\$1,746
Student-*t*	99%	\$1,932	\$4,117

***Table 6**: In-sample VaR and ES (portfolio = \$15,000 notional)*

The fitted Student-*t*, at 1.87 degrees of freedom, is low enough that the distribution's theoretical variance is infinite. At 95%, the three models' VaR estimates sit within \$41 of one another; at 99%, however, they diverge sharply and further still for ES, where the Gaussian's \$1,322 is nearly a third of the Student-*t*'s \$4,117. This is consistent with the high-kurtosis pattern documented for the individual stocks.

Each model's in-sample VaR can then be held fixed and tested against the 251 trading days of 2022 (out-of-sample year) not to mention one driven largely by a different kind of stress (rising interest rates rather than the pandemic onset). Ideally, a well-calibrated model's exceedance rate over 2022 should match its nominal level. This is tested formally with Kupiec's (1995) proportion-of-failures test, a likelihood-ratio test of whether the observed exceedance count is statistically consistent with the nominal rate, borrowed here from Zhang and Nadarajah's (2017) review of VaR backtesting methods; note the documented limitation below.

Table 7 shows that all three methods fail about just as badly (almost five times as many exceedances as the nominal frequency) regardless of distribution, using 95% confidence level. At 99%, tail shape actually matters because Student-*t* avoids rejection—extreme days are what a heavy-tailed distribution can capture but a Gaussian cannot. ES lacks as explicit an exceedance-based test as VaR, so its own calibration is checked by comparing what was lost on average to the model's predicted ES (on days VaR was met or exceeded). Empirical ES is accurate to within 1% at 95% and understates realized severity by 7% at 99%. Normal ES understates realized severity by 27% at 95% and by 18% at 99%. Student-*t*'s ES, by contrast, overstates realized severity by 27% at 95% and by 81% at 99%.

model	α	VaR	exceedances	expected	Kupiec p	realized avg. loss	model ES
Empirical	95%	$764.28	60	12.6	~0.0000	$1,339.77	$1,328.53
Empirical	99%	$1,762.79	9	2.5	0.0014	$2,149.83	$1,997.58
Normal	95%	$798.38	56	12.6	~0.0000	$1,379.55	$1,012.61
Normal	99%	$1,147.78	36	2.5	~0.0000	$1,613.59	$1,321.51
Student-*t*	95%	$805.37	56	12.6	~0.0000	$1,379.55	$1,745.86
Student-*t*	99%	$1,931.67	6	2.5	0.0604	$2,273.09	$4,116.60

***Table** 7: 2022 backtest, exceedances, Kupiec's test, and realized severity*

Zhang and Nadarajah (2017) note that this test has limited power for a single year of data. Because it only counts violations without regard to when they occur, the test may fail to reject a model with clustered exceedances. Plotting the 2022 daily losses against each threshold (**Figure 8**) shows that the 95% exceedances concentrate heavily in the first several months, consistent with the broader early-2022 sell-off in growth and technology stocks driven by rising interest-rate expectations. NFLX's two largest moves in the entire sample (see **Table 2**) occurred during this period—given how extreme those two days were just on their own, they plausibly account for a disproportionate share of the early exceedances. Hence, the normal model failed at both levels in frequency and severity of risk measure due to the necessity of fatter tails. The empirical model, while a better performer in estimating severity, lacks, by construction, the capacity to predict the volatility of 2022 not exhibited between 2019 and 2021. The Student-*t* model passes the 99% frequency test at the cost of overestimating severity. Diversification across as many as fifteen stocks spanning numerous sectors did not, in effect, make the portfolio's risk straightforward to model, warranting further analysis of how dependent the stocks are on one another.

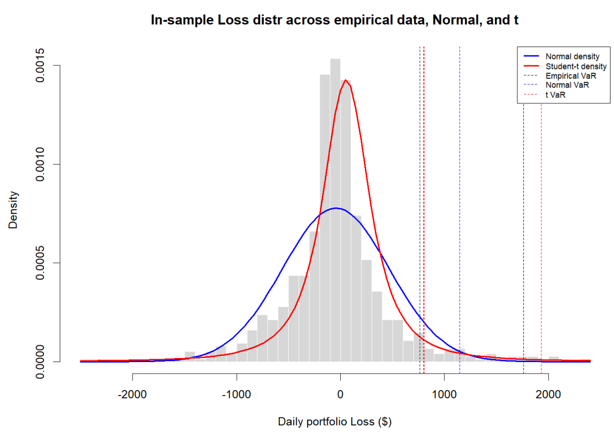


Figure 8

Dependence and Copulas

Gumbel and Clayton are inherently bivariate, so fitting either across all fifteen simultaneously would force every pair to share one dependence parameter, which doesn't work given how unevenly related the fifteen stocks have turned out to be. Gauss and *t* are thus examined across the full sample, all four families on the same illustrative pairs used throughout. At the 15-dimensional level, correlation was estimated via Kendall's rank correlation rather than Pearson, for the same robustness reasons already established. The *t*-copula fits decisively better than Gaussian (indeed, log-likelihood of 3,484.5 against 3,141.3, AIC (how well a model fits historical data that also penalizes overfitting) advantage of 684.22 ($AIC_{Gauss} - AIC_t$ = -6,072.68 - -6,756.90), confirmed by an LR test ($p < 0.001$), the *t*-copula fitted with 7.61 degrees of freedom.

Pair	Best fit	AIC adv. over Gauss	empirical tail comov.	simulated tail comov.
MSFT/NVDA	*t*	29.02	6.4x	5.2x
PFE/REGN	*t*	35.47	3.3x	3.1x
UAL/WMT	*t*	58.05	2.1x	2.3x
TSLA/DUK	*t*	24.57	1.7x	1.6x

***Table** 8: Pairwise copula fit and tail-dependence*

The *t*-copula wins for all pairs. Here, tail comovement measures how frequently a pair's worst 10% of days coincide expressed as a multiple of the 0.1 rate chance would produce; the two right-hand columns compare this measured directly from the data against what each pair's winning fitted copula predicts. The chosen model reproduces real crash-day behavior in three of the four pairs, MSFT/NVDA the exception. Interestingly, the strongest-dependence pair in the set (6.4x) is also where even the best-fitting model understates what really happened. Gumbel captures dependence specifically in a distribution's upper tail, Clayton lower; MSFT/NVDA, the only pair with both fitted $\theta = 1.93$ and $\theta = 1.66$, produces upper- and lower-tail dependence coefficients of 0.57 and 0.66, respectively, so only one direction is in fact described. The *t*-copula's tail dependence is symmetric, and the data does have extremes on both sides (e.g., **Table 2**'s SPG entry being a joint rally on vaccine news) which is a substantive reason the symmetric model wins. This portfolio under a Gaussian framework (that is, zero tail dependence) would predict that stocks weakly related on an ordinary day remain so on the worst of days. The pairs, though, exhibit measured tail comovement well above chance. This parallels the previous section's fitted portfolio loss v of 1.87, namely in that diversifying $1,000 across fifteen stocks in a plethora of sectors provides genuine protection against any single stock's *idiosyncratic* bad day yet comparatively little against their shared tendency to fall in unison during a crash.

Every section revealed that the Normal model doesn't properly describe how the fifteen stocks behave, particularly when the market faces a crisis, neither individually, jointly, nor aggregated into a portfolio. Moreover, in light of the 2022 backtest showing even the best-fitting fat-tail model overstated realized severity, no alternative was perfect. The copula analysis demonstrated that the stocks' interrelations follow a fat-tailed, symmetric dependence structure. Diversification across sectors provides real protection against a given stock's idiosyncratic drop though less against stocks' tendency to crash together. No single framework was fully reliable, thus underscoring the necessity to test on held-out extreme events in choosing and calibrating models most conducive to risk mitigation.

Appendix

Zhang, Y. and Nadarajah, S. (2017). "A review of backtesting for value at risk", *Communications in Statistics - Theory and Methods*, pp. 1–24. doi: https://doi.org/10.1080/03610926.2017.1361984.

Gemini AI was used to plot results, format printed output, and assist in certain miscellaneous, though far from all, coding tests. Please find the report's full coding repository here, where comments indicate wherefore Gemini was used. Gemini AI was also used to recommend including in the report Kupiec's test as a measure of accuracy of a given model's VaR. Lastly, Gemini AI was used to suggest which parts of the full code to include in the report—not everything can be included here due to the imposed page limit.

Comments and diagnostic output are not included here for substantive methodology and checks in order to satisfy the eight-page limit. A full repository is available on the GitHub hyperlink immediately above.

Please note that to remain within the prescribed page limit, the scope of this report has been constrained and may not include sufficiently rigorous justification of its results; a more exhaustive analysis would better capture the topic's complexity, provide the explanations it warrants, and demonstrate a more thorough understanding of the course material. Regrettably, any unconventional formatting throughout this document is a necessary consequence of balancing these arbitrary space restrictions with the mandate to present a complete report.